NumPy Introduction (Detailed Guide for PDF)

What is NumPy?

NumPy (Numerical Python) is a powerful Python library used for numerical computing. It provides support for large, multi-dimensional arrays and matrices along with a collection of mathematical functions.

Why Use NumPy?

- Faster than Python lists
- Efficient memory usage
- Supports mathematical operations
- Used in data science and machine learning

Features of NumPy

- N-dimensional array (ndarray)
- Mathematical functions
- Broadcasting
- Linear algebra operations

Installing NumPy

```
pip install numpy
```

Importing NumPy

```
import numpy as np
```

Creating Arrays

```
import numpy as np
```

```
arr = np.array([1, 2, 3])
```

Types of Arrays

1D Array

```
np.array([1,2,3])
```

2D Array

```
np.array([[1,2],[3,4]])
```

Array Properties

```
arr.shape
arr.size
arr.dtype
```

Array Operations

```
arr + 2
arr * 2
arr - 1
```

Indexing and Slicing

```
arr[0]
arr[1:3]
```

Mathematical Functions

```
np.sum(arr)
np.mean(arr)
np.max(arr)
np.min(arr)
```

Reshaping Arrays

```
arr.reshape(2,2)
```

Special Arrays

```
np.zeros((2,2))
np.ones((2,2))
np.eye(3)
```

Random Numbers

```
np.random.rand(3)
np.random.randint(1,10,3)
```

Broadcasting

Allows operations on arrays of different shapes.

```
arr + 5
```

Linear Algebra

```
np.dot(arr, arr)
```

Example

```
import numpy as np

a = np.array([1,2,3])
b = np.array([4,5,6])

print(a + b)
```

Advantages of NumPy

- High performance
- Easy to use
- Supports large data

Disadvantages of NumPy

- Requires understanding of arrays
- Less flexible than lists

Best Practices

- Use vectorized operations
- Avoid loops
- Use built-in functions